

Nifty 100 Financial Intelligence Platform

Analyst Guide

This guide covers how to use the Screener, Peer Comparison, and other dashboard screens, plus the REST API, to analyze the 92 companies in this platform. It also documents a known data-quality issue that affects a small number of companies -- read the Data Quality Caveats section before drawing conclusions from ROE, ROCE, D/E, or Asset Turnover for any company.

1. Getting Started

The platform has three ways to access data: the Streamlit dashboard (for interactive browsing), the REST API (for programmatic access), and the raw SQLite database at data/nifty100.db (for direct SQL queries or exports to Excel/Python).

To start the dashboard: `streamlit run src/dashboard/app.py`, then open <http://localhost:8501>.

To start the API: `uvicorn src.api.main:app --port 8000`, then browse

<http://localhost:8000/docs> for interactive API documentation.

Coverage note: the platform tracks 92 companies, not the full 100 implied by "Nifty 100" -- 8 tickers present in the raw source files were found to be genuinely absent from the master company reference file during data cleaning, and were excluded consistently across every table rather than left as orphaned records with no company profile.

2. Dashboard Screens

Overview (Home)

Summary KPIs (median ROE, median P/E), sector distribution, ROE distribution histogram, and a market-health banner. ROE is shown as a median, not a mean -- a handful of companies with unusually extreme ROE values would otherwise distort a simple average (see Section 6).

Company Profile

Search any of the 92 companies by ticker or name. Shows KPI tiles, 10-year P&L;/balance sheet/cash flow charts, and pros/cons. If a company's data carries a data-quality caveat for its latest year, a warning banner appears above the KPI tiles -- read it before citing the numbers below it.

Screener

Filter companies by ROE, D/E, FCF, P/E, P/B, dividend yield, CAGR thresholds, and sector. Results below the filter sliders are highlighted in yellow if they carry a data-quality caveat. Results can be downloaded as CSV.

Peer Comparison

Select one of 11 peer groups to see an 8-axis radar chart (ROE, ROCE, NPM, D/E, FCF, and 3 CAGR metrics) and a side-by-side percentile table. Not every company belongs to a peer group -- coverage is 56 of 92 companies.

Trend Analysis

10-year sparkline for any single P&L;/balance-sheet metric per company, with year-over-year % change, or an overlay of multiple metrics at once.

Sector Analysis

Bubble chart (revenue vs. ROE, sized by market cap) and sector median KPI comparison. Extreme ROE outliers are excluded from the bubble chart display for readability, with a note showing how many were excluded.

Capital Allocation Map

Treemap of all companies by their latest year's cash-flow sign pattern (8 categories, e.g. "Reinvestor", "Distress"). Click any segment to drill into the underlying company list.

Annual Reports

Links to BSE-hosted annual report PDFs per company/year, where available. Link presence does not guarantee the link currently resolves -- see Section 5.

Valuation

P/E, P/B, EV/EBITDA, dividend yield, and FCF yield per company, with an overvaluation flag.

IMPORTANT: market cap and valuation multiples in this dataset are SIMULATED, not real market prices -- see Section 7.

3. Screener Guide

The screener applies threshold filters defined in `config/screener_config.yaml` to the latest reported year of every company. Six presets are built in:

Preset	Criteria
Quality	ROE > 15%, D/E < 1.0, FCF > 0
Value	P/E < 20x, P/B < 3x
Growth	5yr PAT CAGR > 20%
Dividend	Dividend yield > 2%
Momentum	5yr Revenue CAGR > 15%
Debt-Free	Borrowings = 0

Debt/Equity filters automatically exclude financial-sector companies (banks, NBFCs, insurers) from the D/E threshold -- their business model runs on leverage, so a D/E<1 "quality" bar would disqualify every bank rather than say anything meaningful about quality. This carve-out is applied consistently, not just for the Quality preset.

Custom filters: any threshold in `screener_config.yaml` can be adjusted without touching code. Add a new named preset by adding a new key under the `presets:` section and selecting which thresholds it references.

4. Peer Comparison Guide

Percentile ranks are computed within each peer group only -- a company's 80th-percentile ROE among 5 IT Services peers is not comparable to an 80th percentile among 23 Financials peers. D/E is inverted in percentile scoring (lower D/E scores higher), since less leverage is generally the more favorable position, unlike every other axis where higher is better.

"Toggle between years" is not available on this screen -- peer percentiles are computed for each company's latest reported year only, not a historical series. Companies not assigned to any peer group (36 of 92) still appear on the dashboard's radar charts (reports/radar_charts/), benchmarked against the full 92-company universe instead of a specific peer group.

5. Valuation Guide

Overvaluation is flagged when at least 2 of 3 valuation multiples exceed their spec-defined "fair" upper bound: $P/E > 25x$, $P/B > 3x$, $EV/EBITDA > 18x$. Because this dataset's simulated market cap data has a median P/E of $\sim 46x$ and median P/B of $\sim 7.5x$ -- both already above those "fair" thresholds -- a large majority of companies (roughly 80-90%) will trip this flag. This reflects the simulated dataset's own central tendency, not a real signal that most of the Nifty 100 is mispriced. Use the flag as a relative screen within this dataset, not as investment advice.

6. Data Quality Caveats -- Read Before Using ROE, ROCE, D/E, or Asset Turnover

Four companies -- **BEL, HAL, INDIGO, and LT** -- have balance sheet figures (total assets, equity, reserves, borrowings) that appear scaled roughly 100x too small relative to their P&L; for certain years. This was found by manually cross-checking ROCE values during Sprint 2 and confirmed against each company's Sales/Total Assets ratio, which reaches into the hundreds (INDIGO: 354x in FY2013; a normal company like TCS is ~1.5x).

This was NOT corrected in the underlying data -- there was no reliable way to determine the true scale factor without the original source workbook, and guessing wrong would have injected fabricated numbers into real financial data. Instead, every affected company-year is flagged:

- The exact affected (company, year) pairs are listed in `output/scale_anomaly_flags.csv` and `output/ratio_edge_cases.log`.
- The dashboard's Company Profile, Screener, Peer Comparison, and Valuation screens all show a warning banner or highlighted row for affected companies.
- The API's `/companies/{ticker}/ratios`, `/screener`, `/sectors/{sector}/companies`, and `/peers` endpoints attach a `data_quality_caveat` field to affected rows.
- `cluster_labels.csv` marks affected companies' cluster assignment as UNRELIABLE, since ROE and D/E are 2 of the 5 KMeans clustering inputs.

Not affected: every other company, and every other year for these 4 companies (e.g. HAL's 2013-2015 data is clean; only 2016 onward is flagged). Other metrics not derived from the balance sheet (e.g. Net Profit Margin, EPS) are unaffected even in flagged years.

7. Other Known Limitations

- `market_cap.xlsx` and `stock_prices.xlsx` are SIMULATED data (per the original project spec), not real market prices -- every valuation multiple and price chart in this platform is a demonstration of the methodology, not real financial information.
- Annual report links (`documents.xlsx`) are shown by presence only -- a link being on file does not guarantee it currently resolves; automated URL validation is available (`src/etl/validator.py`, DQ-13) but disabled by default since it's slow.
- `analysis.xlsx` and `prosandcons.xlsx` have partial coverage (~9-17% of companies) -- most companies will show "no data available" on those specific fields.
- CAGR figures (Revenue, PAT, EPS) are computed on demand from the underlying P&L; data, not stored in the database -- they will not appear in raw SQL queries against `financial_ratios`.

8. API Quick Reference

Base URL: <http://localhost:8000/api/v1>. Interactive documentation (Swagger UI): <http://localhost:8000/docs>. All 16 endpoints return JSON.

Endpoint	Purpose
GET /companies	List/search companies, filter by sector
GET /companies/{ticker}	Full company profile + latest KPIs
GET /companies/{ticker}/pl bs cashflow	Raw financial statement history
GET /companies/{ticker}/ratios	Computed KPI history, with caveat flags
GET /companies/{ticker}/tearsheet	PDF tearsheet download
GET /companies/{ticker}/documents	Annual report links
GET /companies/{ticker}/peers/compare	Radar data vs. peer group average
GET /screener	Filtered company list by threshold params
GET /sectors	Sector-level median KPIs
GET /sectors/{sector}/companies	Companies in one sector
GET /peers/{group_name}	All members of a peer group with percentiles
GET /market-cap/{ticker}	Valuation multiples (SIMULATED data)
GET /portfolio/stats	P10-P90 percentile table, 10 KPIs
GET /health	Service status, DB row counts, uptime

9. Common Analyst Workflows

Workflow A -- Find quality companies undervalued relative to the sector

- Open the Screener screen, enable the ROE and D/E sliders, set $ROE \geq 15\%$ and $D/E \leq 1.0$.
- Sort the results table by P/E ascending (click the column header) to surface the cheapest names within that quality filter.
- Cross-check any candidate on the Sector Analysis screen -- compare its P/E against the sector median bar chart before concluding it's cheap in absolute terms.
- Open the candidate's Company Profile to check the pros/cons and confirm there's no data-quality caveat banner before relying on its ROE.

Workflow B -- Compare a company against its direct peers

- Open Peer Comparison, select the company's peer group from the dropdown.
- Read the radar chart: axes further from center = higher percentile within the group on that metric. The dashed line at the midpoint represents the group average by definition.
- Use the side-by-side table below the chart for exact values rather than estimating from the radar shape.
- If the company isn't in the dropdown's member list at all, it has no peer group assigned (56 of 92 companies do) -- use Sector Analysis instead for a broader comparison.

Workflow C -- Investigate one company in depth via the API

- `GET /api/v1/companies/{ticker}` for the profile + latest KPIs in one call.
- `GET /api/v1/companies/{ticker}/ratios` for the full multi-year KPI history -- check every row's `data_quality_caveat` field before using ROCE/D-E/Asset Turnover from any specific year.
- `GET /api/v1/companies/{ticker}/peers/compare` for a machine-readable version of the radar chart data.
- `GET /api/v1/companies/{ticker}/tearsheet` to download the 2-page PDF summary for offline review.

10. KPI Glossary

Formulas and interpretation for the KPIs used throughout the platform. "None" in the Edge Case column means the platform stores a null value rather than a computed number when that condition is met, so the dashboard/API can distinguish "cannot be computed" from "computed as zero".

Metric	Formula	Edge Case
Net Profit Margin (NPM)	$\text{net_profit} / \text{sales} \times 100$	None if sales = 0
Operating Profit Margin (OPM)	$\text{operating_profit} / \text{sales} \times 100$	None if sales = 0
Return on Equity (ROE)	$\text{net_profit} / (\text{equity} + \text{reserves}) \times 100$	None if equity+reserves $\leq$ 0
Return on Capital (ROCE)	$\text{EBIT} / (\text{equity} + \text{reserves} + \text{borrowings}) \times 100$	Sector-relative for banks/NBFCs
Debt-to-Equity (D/E)	$\text{borrowings} / (\text{equity} + \text{reserves})$	0 = debt-free; financials excluded from >5 flag
Interest Coverage (ICR)	$(\text{op_profit} + \text{other_income}) / \text{interest}$	999 sentinel = debt-free (interest=0)
Asset Turnover	$\text{sales} / \text{total_assets}$	None if total_assets = 0
Free Cash Flow (FCF)	$\text{operating_activity} + \text{investing_activity}$	Negative allowed
Revenue/PAT/EPS CAGR	$(\text{end}/\text{start})^{(1/n)} - 1, n \in \{3,5,10\}\text{yr}$	Turnaround flag if base $\leq$ 0, end > 0
Book Value / Share	$(\text{equity} + \text{reserves}) / (\text{equity_capital}/\text{face_value})$	None if face_value missing/zero
CFO Quality Score	$\text{operating_activity} / \text{net_profit}$	>1.0 = high quality; <0.5 = accrual risk
CapEx Intensity	$\text{abs}(\text{investing_activity}) / \text{sales} \times 100$	>8% = capital intensive
FCF Yield	$\text{FCF} / \text{market_cap_crore} \times 100$	Requires market_cap (SIMULATED data)

Capital allocation patterns (Capital Allocation Map screen): classified by the sign of (CFO, CFI, CFF) for each company's latest year. "Reinvestor" and "Shareholder Returns" share the same (+,-,-) sign pattern and are distinguished by CFO/PAT quality; "Distress" is (-,?,+) -- negative operating cash flow covered by raising external funds. The remaining 5 of the 8 possible sign combinations are labelled using standard capital-allocation terminology, since the source specification only names 3 of the 8 explicitly.

11. Troubleshooting

Q: A metric shows "N/A" for a company I know reports it.

A: Check whether the underlying source value was actually zero, negative, or missing for that company-year -- most KPIs here deliberately show None/N/A instead of a computed number in edge cases (e.g. ROE when equity is negative, NPM when sales is zero) rather than returning a misleading value. See the KPI Glossary (Section 10) for the specific edge-case rule per metric.

Q: A company doesn't appear in the Peer Comparison dropdown for a sector I'd expect.

A: Peer group membership (peer_groups.xlsx) is a separate, manually-curated dataset from sector classification (sectors.xlsx) and covers only 56 of 92 companies. A company can have a sector but no peer group, or vice versa is not possible since peer groups reference specific tickers.

Q: The Valuation screen flags almost every company as overvalued.

A: This is expected given the simulated market_cap data's central tendency -- see Section 5 (Valuation Guide) for the full explanation. It is not a bug and not a real market signal.

Q: A ratio for BEL, HAL, INDIGO, or LT looks implausibly large (e.g. ROE in the thousands of percent).

A: This is the known scale anomaly documented in Section 6. Check the data_quality_caveat field (API) or warning banner (dashboard) before using that specific company-year's ROCE, D/E, Asset Turnover, or Book Value/Share.

Q: The dashboard or API is slow on first load.

A: Both cache expensive computations (screener universe, CAGR calculations) after the first request -- subsequent requests within the cache window (10 minutes) are fast. The very first request after starting either service will be slower while the cache warms.